

RITIKA RAJENDRA RAWAT

Computational Biology | Evolutionary Cancer Genomics | Genomic Variation

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RESEARCH PROFILE

Computational biologist investigating how evolutionary constraint and genomic variation can prioritize functionally important cancer-associated sites. MSc research combines mammalian TP53 comparative genomics with residue-level analysis and statistical testing; current work extends this methodological focus to benchmarked short-read variant analysis. Long-term direction: computational prioritization, biological interpretation, and experimentally testable hypotheses, without treating computational evidence as functional validation.

EDUCATION

MSc Bioinformatics | Guru Nanak Khalsa College, University of Mumbai | Mumbai, India | Jul 2024 - Apr 2026

Academic result: CGPA 8.22/10

Primary research: Evolutionary conservation and functional constraint of TP53 mutation hotspots across mammalian species.

B.Voc. Pharma Analytical Sciences | Ramnarain Ruia Autonomous College | Mumbai, India | Sep 2021 - Apr 2024

Academic result: CGPA 8.57/10

SELECTED RESEARCH OUTPUTS

1. Rawat, R. R.; Nadar, S.; Uppal, G. K. *Evolutionary Conservation and Functional Constraint of TP53 Mutation Hotspots Across Mammalian Species*. Research Square, Version 1, 2026. **First author; preprint**. doi:10.21203/rs.3.rs-9299199/v1
2. Rawat, R. R.; Nadar, S.; Uppal, G. K. *Comparative In-Silico Mapping of TP53 Mutation Hotspots in Elephants: A Responsible Bioinformatics Innovation Contributing to Cancer Research*. In: Ideathon on Vikshit Bharat: Ideas, Innovation and Impact. RED'SHINE Publication, Stockholm, 2026. **First author; published book chapter**. doi:10.25215/9141002199

ACCEPTED CONFERENCE CONTRIBUTION

Accepted Poster Presentation - First and Presenting Author | 25th European Conference on Computational Biology (ECCB 2026) | Geneva, Switzerland | 31 Aug - 4 Sep 2026

Poster: C-G.32, Poster Session 3 - Evolutionary Conservation and Functional Constraints of TP53 Mutation Hotspots Across Mammalian Species.

Co-authors: Sermarani Nadar and Gursimran Kaur Uppal.

Additional engagement: Accepted ECCB 2026 volunteer; scheduled for 31 Aug and 1 Sep 2026, with the specific operational assignment pending.

SELECTED RESEARCH EXPERIENCE

Evolutionary Conservation and Functional Constraint of TP53 Mutation Hotspots Across Mammalian Species | MSc Research Project, Guru Nanak Khalsa College, University of Mumbai | Nov 2025 - Apr 2026

- a. Expanded the original 10-species preprint analysis to 56 mammalian TP53 sequences, with accession-level curation and human-reference coordinate mapping.
- b. Mapped six canonical human cancer-associated hotspots (R175, G245, R248, R249, R273, and R282) and quantified residue conservation, sequence variability, gap fraction, and Shannon entropy.
- c. Compared hotspot and control residues using Mann-Whitney U testing and permutation analysis against a defined DNA-binding-domain background.
- d. Observed complete conservation at all six canonical hotspots in the current dataset; the canonical-hotspot comparison yielded an empirical one-sided p-value of 0.0236 under the defined permutation design.
- e. Produced analysis scripts, figures, statistical tables, accession provenance, methods, limitations, and reviewer-oriented documentation; interpreted conservation as prioritization evidence rather than proof of function or cancer resistance.

Repository: github.com/Rita1791/TP53-Evolutionary-Conservation-Mammals

Comparative In-Silico Mapping of TP53 Mutation Hotspots in Humans and Elephants | Research Project, Guru Nanak Khalsa College, University of Mumbai | Jul 2025 - Mar 2026

- a. Curated public human, African elephant, and Asian elephant TP53 protein sequences, including canonical and available retrogene sequences, for protein-level comparative analysis.
- b. Mapped the six canonical human TP53 hotspots through multiple sequence alignment and evaluated residue conservation in canonical elephant sequences and position-specific variation among retrogene copies.
- c. Integrated human mutation frequency, cross-species conservation, and retrogene variability in a descriptive, non-predictive prioritization framework for residues requiring further investigation.
- d. Used MAFFT, Jalview, MEGA, and Python-based processing for alignment inspection, phylogenetic context, tabulation, and interpretation; findings were published as a first-author book chapter.

Quantitative Determination and Antioxidant Potential of Ferulic Acid from Foxtail Millet | Bachelor's Research Project under Faculty Supervision | 2023 - 2024

- Prepared samples through weighing, grinding, filtration, evaporation, and dilution under faculty supervision.
- Performed TLC with UV visualization and Rf calculation, prepared DPPH reagent and standards, conducted serial dilutions, and recorded spectrophotometric measurements at 517 nm.
- Calculated percentage inhibition and IC50 (approximately 78 ug/mL) and maintained laboratory documentation under safety and SOP requirements.

PROFESSIONAL RESEARCH EXPERIENCE

Bioinformatics Research Assistant | Nainsense Labs Private Limited (NainFit.ai / Ctrlznw) | Mumbai, India | Jul 2026 - Present

Progression: Promoted from Bioinformatics Intern (May - Jun 2026) to a full-time research role.

- Designed and scripted a Bash/Python short-read workflow covering FastQC/MultiQC, read preprocessing, alignment, variant calling, VCF normalization, filtering, and structured result generation.
- For regional caller evaluation, benchmarked GIAB HG001/NA12878 calls against GRCh38 chromosome 22 truth data across 50 selected regions using bcftools isec and RTG vcfeval.
- Measured 99.84% precision, 95.10% sensitivity, and 97.41% F-measure within the defined regional benchmark; these metrics are not presented as whole-genome, end-to-end, or clinical validation.
- Examined precision-sensitivity differences, discordant calls, and low-recall regions; generated figures, tables, technical reports, and manuscript-related sections for review and communication.
- Led technical and healthcare-workflow demonstrations for six US-based external stakeholders (one physician, two healthcare organizations, two researchers, and one business client), translating genomics, telehealth, and AI-assisted workflows for non-technical audiences.
- Converted stakeholder feedback into workflow diagrams, client documentation, presentations, and requirements for product, development, and founding teams.

Clinical Trial Assistant Intern | S. P. Mandali's IATRIS | On-site | Apr 2024 - Jun 2024

Scope: Clinical-trial documentation, protocol-aware data handling, regulatory records, and documentation quality control.

COMPUTATIONAL METHODS

Programming and research computing: Python, Bash/Linux, R, Jupyter, Git/GitHub; Biopython, pandas, NumPy, SciPy, and Matplotlib.

Comparative and evolutionary genomics: NCBI/UniProt sequence retrieval, accession curation, protein multiple sequence alignment, human-coordinate residue mapping, conservation scoring, Shannon entropy, and phylogenetic/structural context.

Genomics and variant analysis: FASTQ/BAM/VCF processing; FastQC, MultiQC, fastp, BWA-MEM/BWA-MEM2, SAMtools, bcftools, VCF normalization/filtering, RTG vcfeval, and GIAB benchmarking.

Statistical analysis: Mann-Whitney U testing, permutation testing, empirical null distributions, precision, sensitivity/recall, F-measure, and data visualization.

EXPERIMENTAL AND ANALYTICAL FOUNDATION

Faculty-supervised laboratory experience complementing the computational profile; no current claim of independent experimental validation.

Direct hands-on work: TLC, GC, HPLC, sample preparation, filtration, evaporation, dilution, UV-Vis spectroscopy, DPPH assays, standards/serial dilution, Rf and IC50 calculations, and laboratory documentation.

Additional laboratory exposure: PCR, gel electrophoresis, DNA/RNA extraction, protein estimation, ELISA, microscopy, centrifugation, titration, chromatography, pharmaceutical analysis, and analytical chemistry.

Guided exposure: FT-IR and dissolution testing.

SELECTED TRAINING

- Bioinformatics Resources and Tools Workshop - Swiss Institute of Bioinformatics (SIB), Jun 2026.
- PDBe Tools on GitHub Workshop - EMBL-EBI / Protein Data Bank in Europe (PDBe).
- SPHN Metadata Catalogue and Interoperability Workshop - Swiss Personalized Health Network (SPHN).

LANGUAGES

Languages: English (C1); Marathi and Hindi (native); German and Italian (A2).

REFERENCES

- **Mr. Ashley Nunes** - Professional Reference, Nainsense Labs Private Limited
Reference available upon request.
- **Dr. Gursimran Kaur Uppal** - Academic Reference, Master's Program
Reference available upon request.
- **Dr. Sasikumar Menon** - Academic Reference, Bachelor's Program
Reference available upon request.